

Problem 2(b) — Full Walkthrough

$\exp((\sigma_1(AX))^{10})$ is convex

Goal: Let $\sigma_1(Z)$ denote the largest singular value of a matrix Z . Show that

$$f(X) = \exp\left((\sigma_1(AX))^{10}\right)$$

is convex in X .

Strategy: Write $f(X) = h(g(X))$ with $g(X) = \sigma_1(AX)$ (the “inner” piece) and $h(t) = \exp(t^{10})$ (the “outer” piece). Show g is convex, show h is convex and nondecreasing on the range g actually takes, then prove a composition rule — directly from the definition, the same way we proved the affine-composition rule in 2(a) — that combines the two facts into convexity of f .

Step 1 — Recognize the structure

Define

$$g(X) := \sigma_1(AX), \quad h(t) := \exp(t^{10}).$$

Then $f(X) = h(g(X))$: first apply the linear map $X \mapsto AX$ and take its largest singular value, then feed that number into h .

Step 2 — $\sigma_1(Z)$ is a norm, hence convex

Recall the variational definition of the largest singular value:

$$\sigma_1(Z) = \sup_{\|u\|=\|v\|=1} u^T Z v.$$

Homogeneity: for any scalar α , replacing u with $-u$ doesn’t change the constraint $\|u\| = 1$, so the supremum over u, v is unchanged if we flip the sign of the quantity being maximized. This gives

$$\sigma_1(\alpha Z) = \sup_{\|u\|=\|v\|=1} u^T (\alpha Z) v = |\alpha| \sup_{\|u\|=\|v\|=1} u^T Z v = |\alpha| \sigma_1(Z).$$

Triangle inequality: since a supremum of a sum is at most the sum of the suprema,

$$\sigma_1(Z + W) = \sup_{u,v} u^T (Z + W) v = \sup_{u,v} [u^T Z v + u^T W v] \leq \sup_{u,v} u^T Z v + \sup_{u,v} u^T W v = \sigma_1(Z) + \sigma_1(W).$$

Nonnegativity: $\sigma_1(Z) \geq 0$ always, with equality only when $Z = 0$.

So $\sigma_1(\cdot)$ satisfies all three norm axioms — it *is* a norm (the spectral norm). **Any norm is convex:** for $\theta \in [0, 1]$,

$$\sigma_1(\theta Z + (1 - \theta)W) \leq \sigma_1(\theta Z) + \sigma_1((1 - \theta)W) = \theta \sigma_1(Z) + (1 - \theta) \sigma_1(W),$$

using the triangle inequality first, then homogeneity (valid since $\theta, (1 - \theta) \geq 0$, so $|\theta| = \theta$). This is exactly the definition of convexity. So $\sigma_1(\cdot)$ is convex, and $\sigma_1(Z) \geq 0$ for every Z .

Step 3 — $g(X) = \sigma_1(AX)$ is convex

This is the same “convex outer function precomposed with a linear map” pattern as 2(a) Step 2, just with $b = 0$ (the map $X \mapsto AX$ is linear, a special case of affine). Take any X, Y and $\theta \in [0, 1]$:

$$g(\theta X + (1 - \theta)Y) = \sigma_1(A(\theta X + (1 - \theta)Y)) = \sigma_1(\theta(AX) + (1 - \theta)(AY)) \leq \theta \sigma_1(AX) + (1 - \theta) \sigma_1(AY),$$

using linearity of A in the middle step and convexity of σ_1 (Step 2) in the last. That’s exactly $g(\theta X + (1 - \theta)Y) \leq \theta g(X) + (1 - \theta)g(Y)$: **g is convex**. And since $g(X) = \sigma_1(AX)$ is a norm value, $g(X) \geq 0$ for every X .

Step 4 — $h(t) = \exp(t^{10})$ is convex, and nondecreasing for $t \geq 0$

Differentiate:

$$h'(t) = 10t^9 e^{t^{10}}.$$

For $t \geq 0$: $t^9 \geq 0$ and $e^{t^{10}} > 0$, so $h'(t) \geq 0$ — **h is nondecreasing on $[0, \infty)$** . (For $t < 0$, $t^9 < 0$, so h is actually decreasing there — but we only care about $t \geq 0$, since that’s the only range g ever produces.)

Differentiate again, using the product rule on $h'(t) = 10t^9 e^{t^{10}}$:

$$h''(t) = 90t^8 e^{t^{10}} + 10t^9 \cdot e^{t^{10}} \cdot 10t^9 = e^{t^{10}} (90t^8 + 100t^{18}).$$

Both t^8 and t^{18} are even powers, so they’re ≥ 0 for *every* real t (not just $t \geq 0$), and $e^{t^{10}} > 0$ always. So $h''(t) \geq 0$ for all $t \in \mathbb{R}$: **h is convex everywhere**.

Step 5 — The composition rule, proved directly

Claim: if g is convex with $g(X) \geq 0$ for all X , and h is convex and nondecreasing on $[0, \infty)$, then $f(X) := h(g(X))$ is convex.

Proof: Take any X, Y and $\theta \in [0, 1]$. Let $s := g(X) \geq 0$ and $t := g(Y) \geq 0$.

By convexity of g :

$$g(\theta X + (1 - \theta)Y) \leq \theta s + (1 - \theta)t.$$

Both sides of this inequality are ≥ 0 (the left because g only outputs nonnegative values; the right because it’s a convex combination of the nonnegative numbers s, t). Since h is *nondecreasing* on $[0, \infty)$, applying h to both sides of a true inequality between two nonnegative numbers preserves the inequality direction:

$$h(g(\theta X + (1 - \theta)Y)) \leq h(\theta s + (1 - \theta)t).$$

Now apply convexity of h to the right-hand side:

$$h(\theta s + (1 - \theta)t) \leq \theta h(s) + (1 - \theta) h(t) = \theta h(g(X)) + (1 - \theta) h(g(Y)) = \theta f(X) + (1 - \theta)f(Y).$$

Chaining the two inequalities together:

$$f(\theta X + (1 - \theta)Y) = h(g(\theta X + (1 - \theta)Y)) \leq \theta f(X) + (1 - \theta)f(Y).$$

This is exactly the definition of f being convex. ■

Note where each hypothesis was used: convexity of g produced the first inequality; **nondecreasing**-ness of h was needed to legally push h through that inequality without flipping it; convexity of h produced the second inequality. Drop any one of the three hypotheses and the chain breaks.

Step 6 — Conclude

From Step 3, $g(X) = \sigma_1(AX)$ is convex with $g(X) \geq 0$ always. From Step 4, $h(t) = \exp(t^{10})$ is convex, and nondecreasing on $[0, \infty)$ — exactly the range g produces. The composition rule from Step 5 then applies directly:

$$f(X) = h(g(X)) = \exp((\sigma_1(AX))^{10}) \text{ is convex.} \quad \blacksquare$$